

Description of family solanaceae in semi technical term

Potato or Nightshade family

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- This family includes 85 genera and 2200 species.
 - **Distribution:** Majority of members of this family are distributed in tropical region and also in temperate regions. The largest genus of this family is *Solanum* including 1500 species.
 - **Habitat:** Wild or cultivated, mesophytes or rarely xerophytes.
 - **Vegetative character**
 - **Habit:** Most of the members of this family are annual or perennial erect or climbing herbs, shrubs or trees.
 - **Root:** Tap root branched,
 - **Stem:** Erect, prostrate or rarely climber, herbaceous, or woody, branched, hairy, solid or fistular, pubescent, cylindrical. Sometimes modified into underground tuber.
 - **Leaf:** cauline or ramal, alternate in vegetative region and opposite in floral region, rarely whorled, exstipulate, petiolate or rarely sessile, simple or rarely compound (in tomato), pubescent, entire or lobed or wavy, unicostate, reticulate venation.

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- **Inflorescence:** cymose, usually axillary
 - **Flower:** Ebracteate, pedicellate, complete, bisexual, regular, actinomorphic, pentamerous, hypogynous.
 - **Calyx:** sepals-5, gamosepalous, campanulate or tubular, persistent, hairy, valvate aestivation, inferior, persistent, pubescent.
 - **Corolla:** petals-5, gamopetalous, infundibuliform or tubular, valvate or twisted aestivation, inferior,
 - **Androecium:** stamens-5, polyandrous, epipetalous, alternating to petals, anther dithecal, connate, basifixed, introse, inferior. Anther dithecal, basifixed introse, filament short, simple and cylindrical.

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- **Gynoecium:** Bicarpellary, syncarpous, ovary superior, bilocular, many ovules in each locule, axile placentation, with obliquely placed swollen placenta, style short, stigma bilobed or capitate.
 - Pollination: self or cross
 - **Fruit:** Berry or capsule
 - **Seed:** Albuminous with two cotyledons
 - **Pollination:** Entomophilous
 - **Floral formula:** Ebr

Classification

K- Plantae

D – Angiospermae

C- Dicotyledonae

O-Polemoniales

F- Solanaceae

G- Solanum

S- nigrum

Common name –Black night shade plant

Local name – Kali gedi

Diagnostic feature

Pubescent plant body.

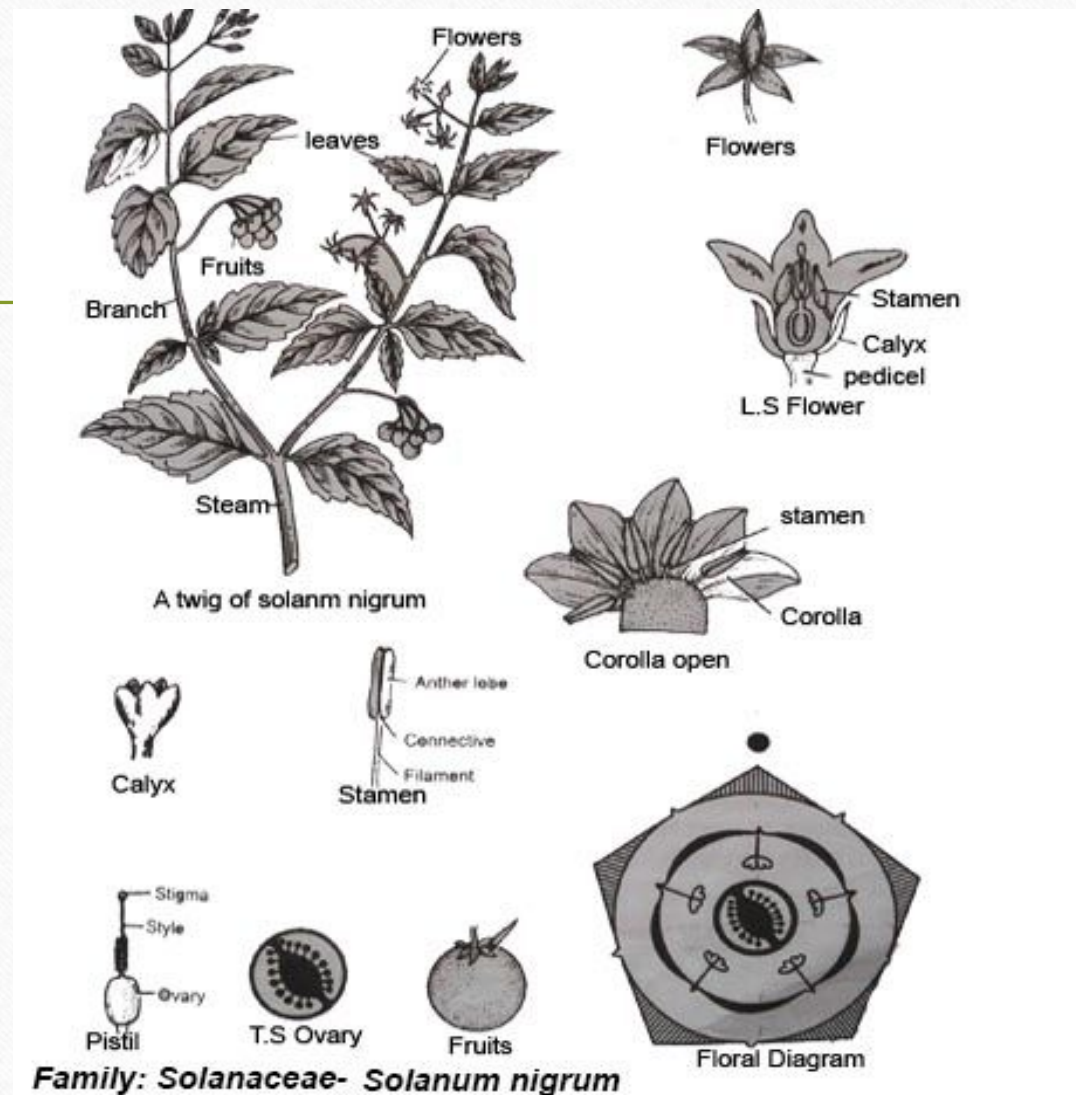
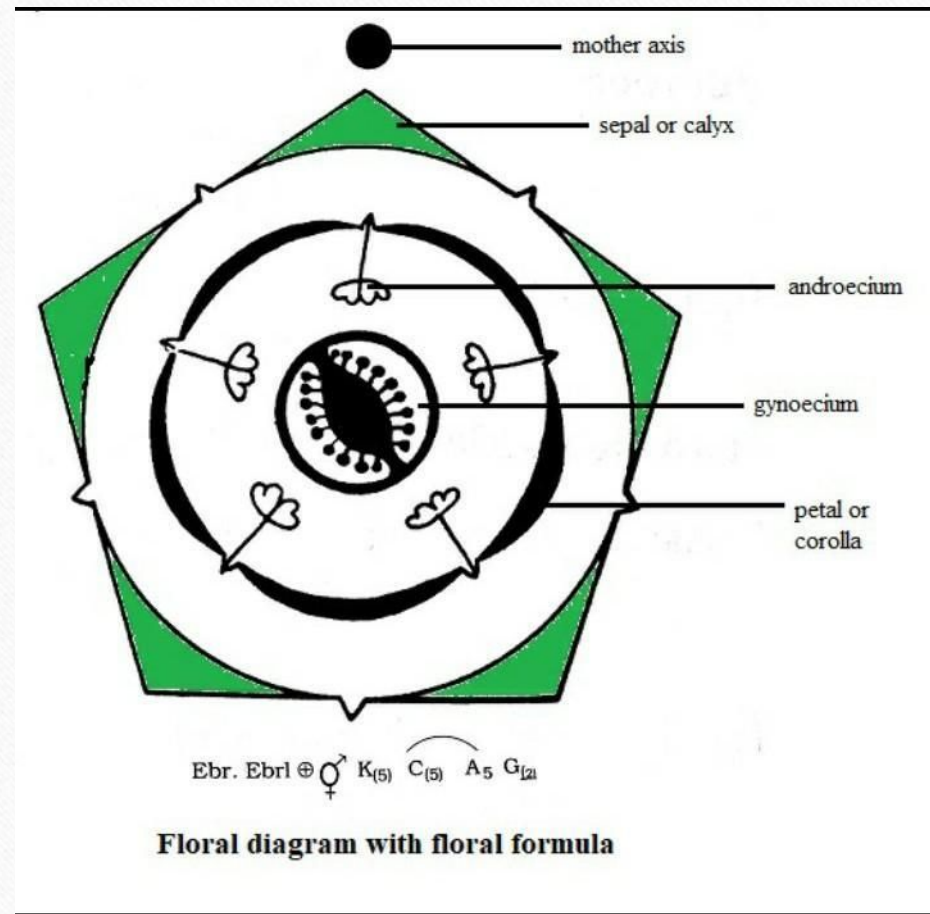
Stem modified in tuber.

Cymose inflorescence

Presence of persistent calyx.

Epipetalous stamen. Axile placentation with obliquely placed swollen placenta.

Fruit berry or capsule.





Capsicum annum



Solanum melongena



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Lycopersicum esculentum



Brugmansis



Solanum nigrum

Economic importance

- Used as vegetable. Eg. Potato, tomato *Solanum tuberosum*
- Some are used as edible fruits.eg, Raspberry (*Rubus idaeus*)
- Used in spices. Eg, chilies
- Tobacco plants used in smoking (*Nicotiana tobacum*)
- Some plants used in medicine. Eg. Belladonna. Atropine is extracted from this plants and used for dilating pupil (*Atropa belladonna*)

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- Ornamental plants
 - Night queen *Cestrum nocturnum*
 - Day king raja *Cestrum diurnum*
 - Petunia *Petunia hybrida*

Thank you